The Case for Voluntary Disclosure

Strategic Feature Revelation in Regression and Classification

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Focus: Characterizing *when* voluntary disclosure outperforms mandatory disclosure in **regression** (squared loss) and **classification** (0-1 loss) decision problems.

Motivation

Motivation: Firm Valuation & Strategic Disclosure

Setting: A market model predicts firm value Y from features X, and sets a **disclosure regime**.

The Dilemma: Firms have private preferred values $U \neq Y$. They act strategically to influence the market:

- **Overvalue** ($U > Y$): seeking investment, inflating stock price before sell-off
  $\Rightarrow$ *reveal* strong metrics, hide liabilities.
- **Undervalue** ($U < Y$): management buyout, tax avoidance
  $\Rightarrow$ *conceal* contracts, suppress growth signals.

Two Regimes the Market Can Set

Mandatory

all firms must disclose

vs.

Voluntary

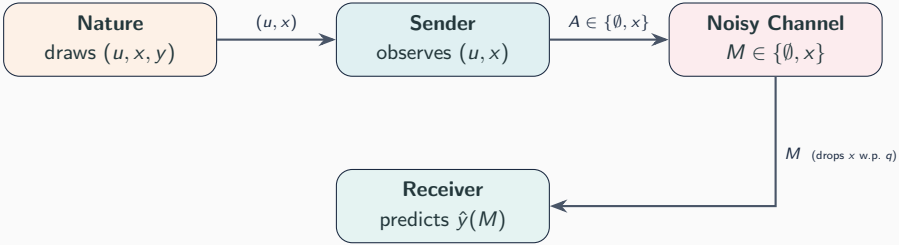
firms *choose* what to reveal

When does voluntary disclosure — driven by self-interest —

outperform *mandatory?*

Problem Formulation

Game-Theoretic Model



Sender loss: $\ell(u, \hat{y})$ | Receiver loss: $\ell(y, \hat{y})$

Sender Strategy

$\mathcal{S} = \{A : \mathcal{Y} \times \mathcal{X} \mapsto \{\emptyset, x\}\}$

Disclose full features x or withhold $\emptyset$

Receiver Strategy

$\mathcal{R} = \{\hat{y} : \{\emptyset, \mathcal{X}\} \mapsto \mathcal{Y}\}$

Predict based on received message M

Stackelberg Equilibrium & Strategic Advantage

We model the interaction as a **screening game** (Stackelberg):

1. **Receiver commits** to a strategy $\hat{y} \in \mathcal{R}$
2. **Sender best-responds:** $A(u, x) = \arg\min_{a \in \{\emptyset, x\}} \mathbb{E}[\ell(u, \hat{y}(M)) \mid A = a]$
3. **Outcome realized** and losses incurred

Strategic Risk (Voluntary)

$$R_S := \min_{\hat{y} \in \mathcal{R}} \mathbb{E}[\ell(Y, \hat{y}(M))]$$

Vanilla Risk (Mandatory)

$$R_V := \mathbb{E}[\ell(Y, \hat{y}_V(M))]$$

Definition: Strategic Advantage

A **strategic advantage** exists if $R_S < R_V$ — voluntary disclosure *strictly outperforms* mandatory.

Regression Results

Regression Setup

- Outcome space: $\mathcal{Y} = \mathbb{R}$, Loss: $\ell(y, \hat{y}) = (y - \hat{y})^2$ (MSE)
- All variables (U, X, Y) have **zero mean**
- Linear receiver: $\hat{y}(\emptyset) = 0$, $\hat{y}(x) = b^\top x$

Vanilla Risk (Mandatory)

$$R_V^\star(q) = \mathrm{Var}(Y) - (1-q)\,\mathrm{Var}(b_Y^\top X)$$

where $b_Y = \mathrm{Var}(X)^{-1}\,\mathrm{Cov}(X, Y)$
is the OLS weight vector.

Strategic Risk (Voluntary)

$$R_S(b, q) = \mathrm{Var}(Y) - (1-q)\mathbb{E}[G \cdot \mathbb{I}\{F > 0\}]$$

Key scalar fields governing strategic behavior:

$$F(x, u) := (2u - b^\top x)\, b^\top x$$ **Sender's decision rule**

$$G(x, y) := (2y - b^\top x)\, b^\top x$$ **Receiver's gain function**

Sufficiency Condition for Regression

Theorem (Sufficiency)

A **strategic advantage** exists if $\exists\, b \in \mathbb{R}^d$ such that:

$$\|U - Y\|_2 < \frac{\mathbb{E}\big[|G(X,Y)|\big] - \mathrm{Var}(b_Y^\top X) - \mathrm{Var}((b - b_Y)^\top X)}{4\sqrt{\mathrm{Var}(b_Y^\top X)}}$$

LHS: $\|U - Y\|_2$

Measures *preference misalignment* between sender's preferred value U and truth Y

RHS: Feature informativeness

Driven by $\mathbb{E}[|G|]$, the expected gain from observing features, minus sub-optimality penalty

Strategic advantage $\Longleftarrow$ **preferences are sufficiently aligned** relative to feature informativeness

Necessary Condition for Regression

Theorem (Necessity)

A **necessary condition** for a strategic advantage to exist:

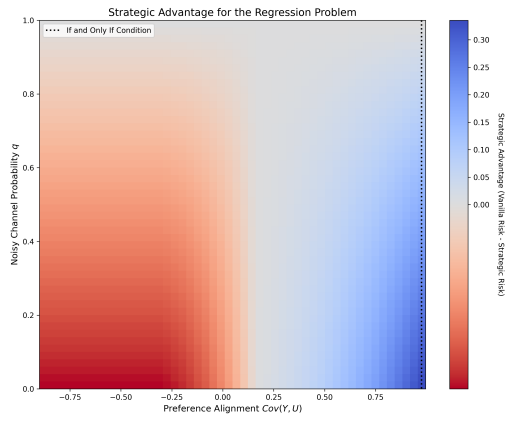
$$\max_b \left(\mathbb{E}\big[|G(X,Y)|\big] - \mathrm{Var}(b_Y^\top X) - \mathrm{Var}((b - b_Y)^\top X) \right) > 0$$

Key Remarks

- Necessary condition ensures sufficiency bound is **non-vacuous**
- ⋆ **Both conditions are independent of channel noise q!**
  q controls the *magnitude* of the advantage, not its *direction*.
- If conditions hold at $b = b_Y$, they hold for optimal b

Takeaway: The noise q determines the **magnitude** but **not the direction** of the strategic advantage.

Regression: Numerical Validation



Setup:

- $(X, Y, U) \sim \mathcal{N}(0, \Sigma)$
- Unit marginal variances
- $\mathrm{Cov}(X, Y) = \mathrm{Cov}(X, U) = 0.2$

Parameter sweeps:

- $q \in [0, 1]$ (channel noise)
- $\mathrm{Cov}(Y, U) \in [-1, 1]$ (preference alignment)

Observation

The **dotted contour** (from our analytical sufficiency bound) reliably delineates the true region of strategic advantage.

Classification Results

Classification Setup

Vanilla Risk

$$R_V^\star(q) = q\,\mathbb{P}(Y = -\operatorname{sign}(\mathbb{E}[Y])) + (1-q)\,\mathbb{E}\left[\frac{1-|\eta(X)|}{2}\right]$$

where $\eta(x) = \mathbb{E}[Y \mid X = x]$.

Strategic Risk

For default option $i \in \{-1, +1\}$:

$$R_{\mathcal{S}_i}^\star(q) = \pi_i\,\mathbb{P}(Y = -i \mid U = i) + \pi_{-i}\left[(1-q)\mathbb{E}\left[\frac{1-|\eta_{-i}|}{2}\right] + q\,\mathbb{P}(Y = -i \mid U = -i)\right]$$

Only **two** possible strategic receiver strategies: default to $+1$ ($\mathcal{S}_1$) or default to -1 ($\mathcal{S}_{-1}$).

The optimal strategy depends on distribution structure and channel noise q.

Classification: Stackelberg Equilibrium

Theorem (Exact Characterization of Strategic Advantage)

Assume $\mathbb{E}[Y] > 0$. We provide an **if and only if** exact condition for strategic advantage. At max noise ($q = 1$), strategy $\mathcal{S}_1$ is always better. Two scenarios arise:

Case 1: $\mathcal{S}_1$ dominates $\forall q$

When $R^\star_{\mathcal{S}_1}(0) \leq R^\star_{\mathcal{S}_{-1}}(0)$:

$$(1-q)\left[\mathbb{E}\left[\frac{1-|\eta|}{2}\right] - \pi_{-1}\mathbb{E}\left[\frac{1-|\eta_{-1}|}{2}\right] - \pi_1\mathbb{P}(Y=-1 \mid U=1)\right] > 0$$

Independent of q! Analogous to regression.

Case 2: Strategy switch at $q^\star$

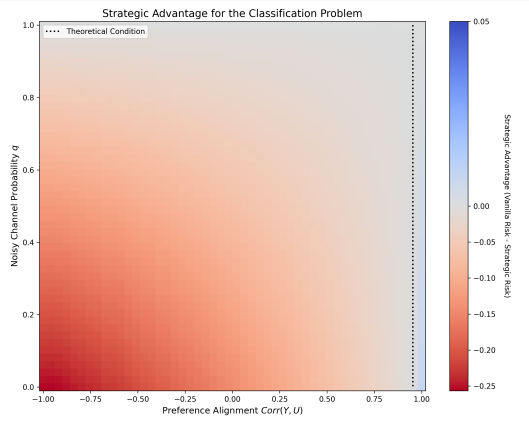
$\exists\, q^\star \in (0,1)$. For $q \geq q^\star$: use $\mathcal{S}_1$.

For $q < q^\star$: use $\mathcal{S}_{-1}$ with:

$$(1-q)\left[\mathbb{E}\left[\frac{1-|\eta|}{2}\right] - \pi_1\mathbb{E}\left[\frac{1-|\eta_1|}{2}\right]\right] + q\left[\mathbb{P}(Y=-1) - \pi_1\mathbb{P}(Y=1 \mid U=1)\right] > \pi_{-1}\mathbb{P}(Y=1 \mid U=-1)$$

Depends on q! Noise affects existence.

Classification: Numerical Validation



Setup:

- $(Y, U) \in \{-1, +1\}^2$
- $P(Y=1) = 0.6,\ P(U=1) = 0.5$
- Features: Gaussian mixture
  $X \mid Y, U \sim Y\mu_Y + U\mu_U + \mathcal{N}(0, 1)$

Parameter sweeps:

- $q \in [0, 1]$ (channel noise)
- $\mathrm{Cov}(Y, U) \in [-1, 1]$ (preference alignment)

Validation

Analytical boundary matches the empirical advantage region precisely.

Conclusion

Summary of Contributions

Regression

Sufficiency & necessity conditions for strategic advantage

Independent of q

Validated numerically

Classification

Full Stackelberg equilibrium characterization

Two-case structure

q-dependent threshold

Key Drivers of Strategic Advantage

Regression: driven by U, Y **alignment.** (Favourable when preferences align with ground truth.)

Classification: driven by the competition between $\mathbb{E}[Y \mid X]$ and the pair $\mathbb{E}[Y \mid X, U], \mathbb{E}[Y]$.

Limitations & Future Directions

Limitations

- Sufficiency bounds can be **conservative** for some distribution families
- Restricted to **linear** receiver strategies in regression
- Classification analysis limited to **binary** labels
- Environment (joint distribution) assumed **known**

Future Directions

- **Unknown environment:** online learning when receiver must learn the distribution
- **Non-linear predictors:** kernels, neural networks
- **Multi-sender** competitive environments
- **Partial disclosure:** reveal subsets of features
- **Tighter bounds:** closing the sufficiency–necessity gap

Thank You!

Questions & Discussion

Key Message

Voluntary disclosure can outperform mandatory disclosure when participant preferences are **sufficiently aligned** *with ground truth.*